Econ 2 - Lecture 10 - 5/4/26

Discussion Activity #3 this week

Lecture Quiz #5 Posted Wednesday

Last Classes:

→ Defined Equilibrium as Y = AE (Output = Spending)

→ Consumption is the largest component of AE

→ Income (Y) is the biggest determinant of C

→ $C = AC + mpc \cdot (Y - T)$ → Fixed Spending + income-driven spending

Real GDP (Y)	Taxes (T)	Disposable Income (Y-T)	Consumption (C)
$\boxed{200}$	100	100	$\boxed{180}$
300	100	200	260
400	100	300	340
500	100	400	420
600	100	500	500
700	100	600	580
800	100	700	660
900	100	800	740

+100 +80

+100 +80

What is mpc? What is AC?

$$mpc = \frac{\Delta C}{\Delta Y-T} = \frac{\Delta C}{\Delta Y} = \frac{260-180}{300-200} = \frac{80}{100} = 0.8$$

What does mpc mean? Fraction of income I spend

What is AC? $C = AC + mpc\,(Y-T)$ (use any C, Y combo)

$C = 180,\ Y = 200$ $\quad 180 = AC + 0.8\,(200 - 100) = AC + 80,\ AC = 100$

$C = 500,\ Y = 600$ $\quad 500 = AC + 0.8\,(600 - 100) = AC + 400$

Incorporate other types of Spenders

Next Group = Firms

Two Major components of firm spending

1. Capital / Machines
2. New Homes

$\Big\}$ Planned Investment Spending (I^P)

Not including unplanned spending
→ Changes in Private Inventories not part of I^P

I^P (Planned Investment Spending) not always equal to I

I^P is assumed to be fixed for all levels of Y
↪ Same as T & AC

Next Group of Spenders → Government!

Government Spending = G
Not including wealth transfers, UE benefits, social security

Goods / Services that are necessary but hard for individual firms to profit from
Assume G is fixed for all levels of Y

Final Group of Spenders: Foreigners
Net Exports (NX) = Exports (X) - Imports (M)
Possible for $NX < 0$, $M > X$

Build the Aggregate Expenditures Equation

$$AE = C + I^P + G + NX$$

What does it mean for $Y = AE$?

$$\underbrace{\cancel{C} + I + \cancel{G} + \cancel{NX}}_{Y} = \underbrace{\cancel{C} + I^P + \cancel{G} + \cancel{NX}}_{AE}$$

$$I = I^P$$

1. Capital	1. Capital
2. New Homes	2. New Home
3. Chg in Inventory	3. $\emptyset$

If Chg in Inventory $= \emptyset$,
$\rightarrow I^P = I \Rightarrow Y = AE$

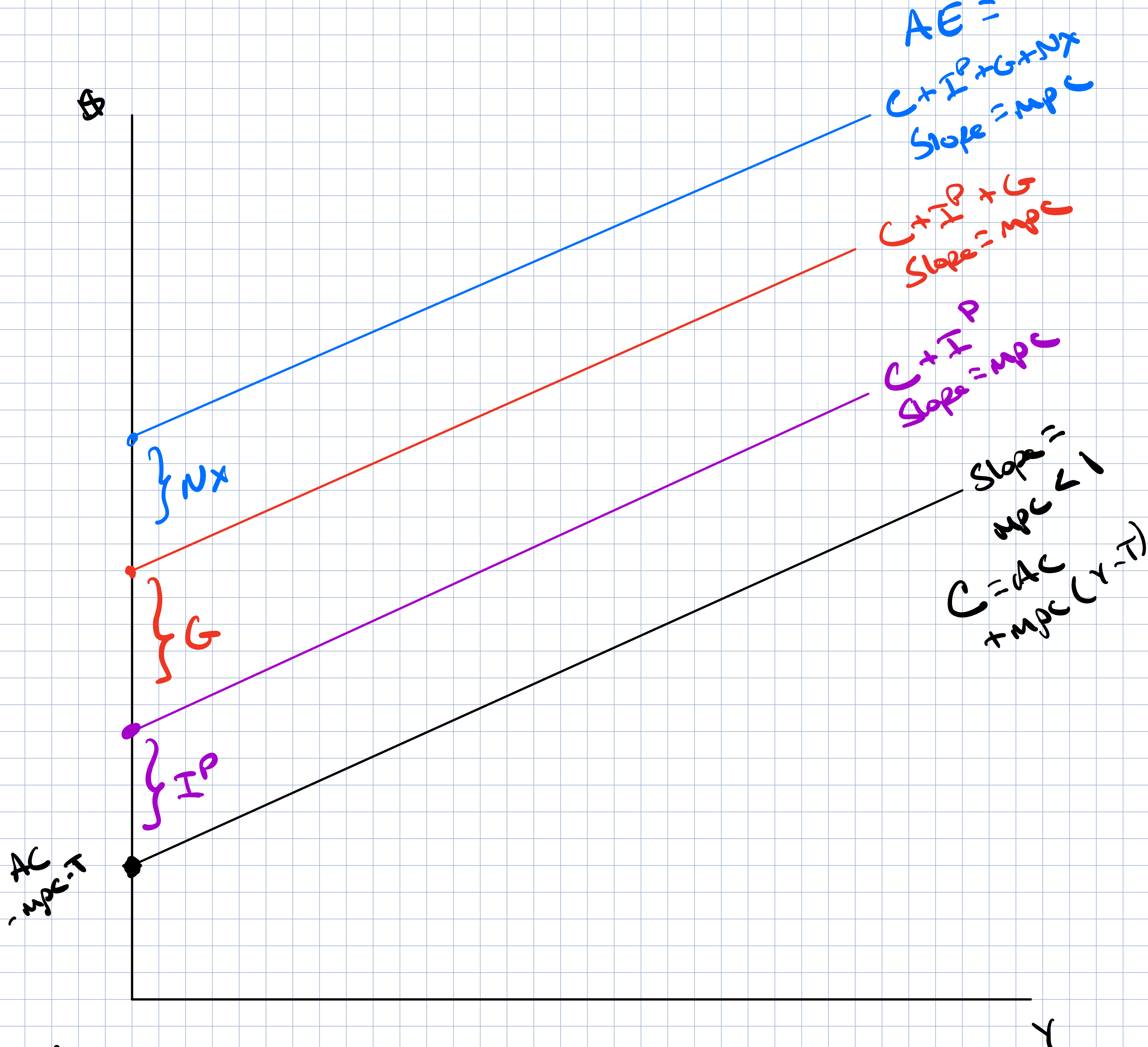
$$\underbrace{\begin{array}{l}\text{If Change in Inventory} = 0 \\ \rightarrow \text{No extra production} \\ \rightarrow \text{No need to go to "stock room" to meet demand} \\ \rightarrow \text{Sell exactly the amount we produce}\end{array}}_{\text{Equilibrium}}$$

Show $Y = AE$ graphically → Start with AE

$$AE = C + I^P + G + NX$$

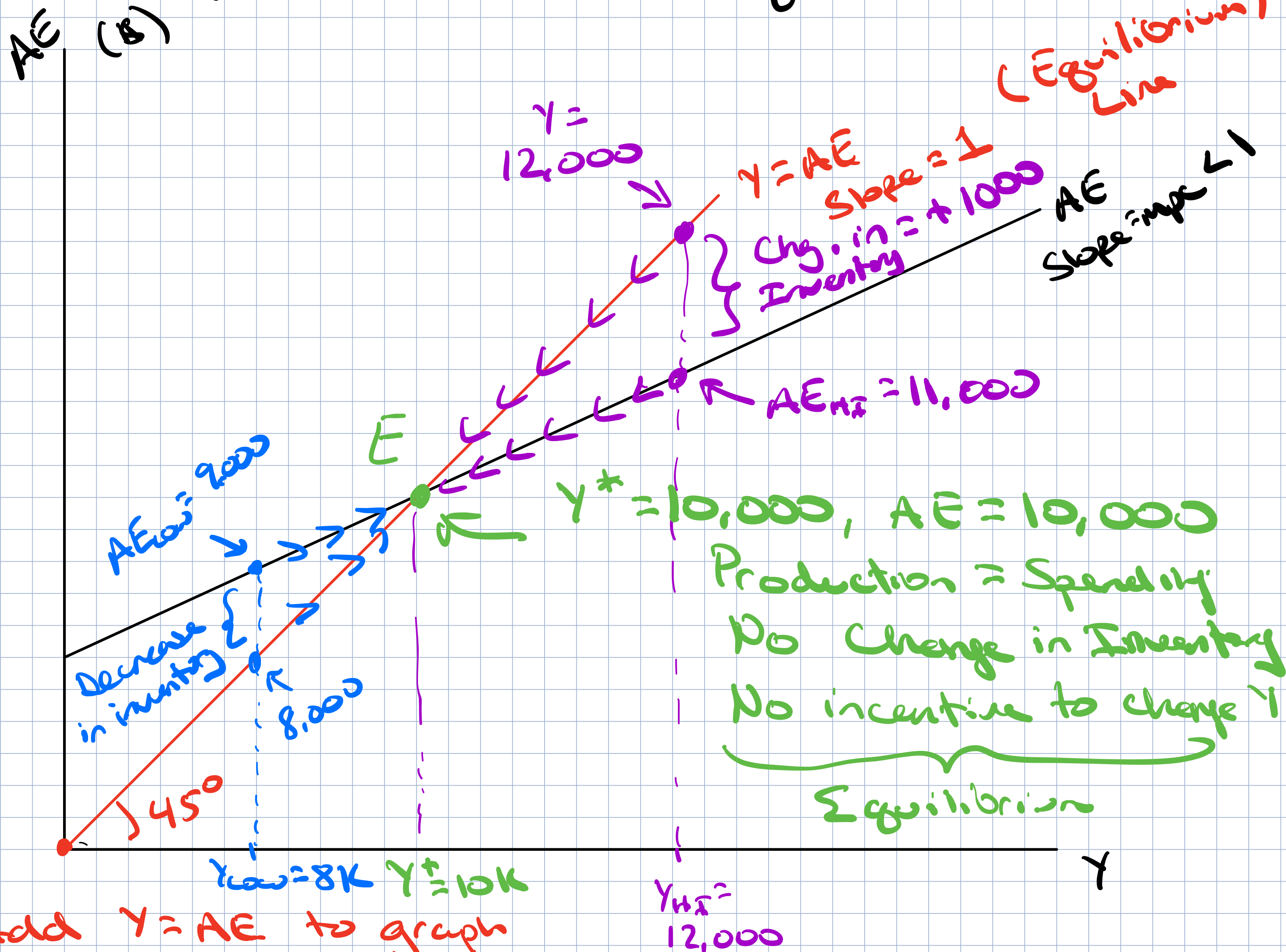
$$= \underset{\text{fixed}}{AC} + \underset{\text{constant}}{mpc} \cdot (Y - \underset{\text{fixed}}{T}) + \underset{\text{fixed}}{I^P} + \underset{\text{fixed}}{G} + \underset{\text{fixed}}{NX}$$

$$AE = \underbrace{\text{fixed value}}_{AC,\ I^P,\ G,\ NX,\ T} + mpc \cdot Y$$



* AE: Captures all components of spending → Shifts in AE → Descriptive

Drawing of AE vs. Y = AE (Equilibrium)



Add Y = AE to graph

If AE = ∅, Y = ∅

If AE = 1000, Y = 1000, AE = 25000, Y = 25000

What happens when $Y \neq AE$?

Start at $Y_{HI} = 12,000$, $AE_{HI} = 11,000$

Production > Spending → Y > AE → Too much Y

Inventory increase by Y − AE = 1,000

Incentive for firms next year? Decrease Y → move along AE

Start at $Y_{LOW} = 8,000$, $AE_{LOW} = 9,000$

Spending > Production, AE > Y, Not enough Y

Decrease in inventory → chg. in inventory = Y − AE

= 8000 − 9000 = −1000

Incentive for firms? Increase Y

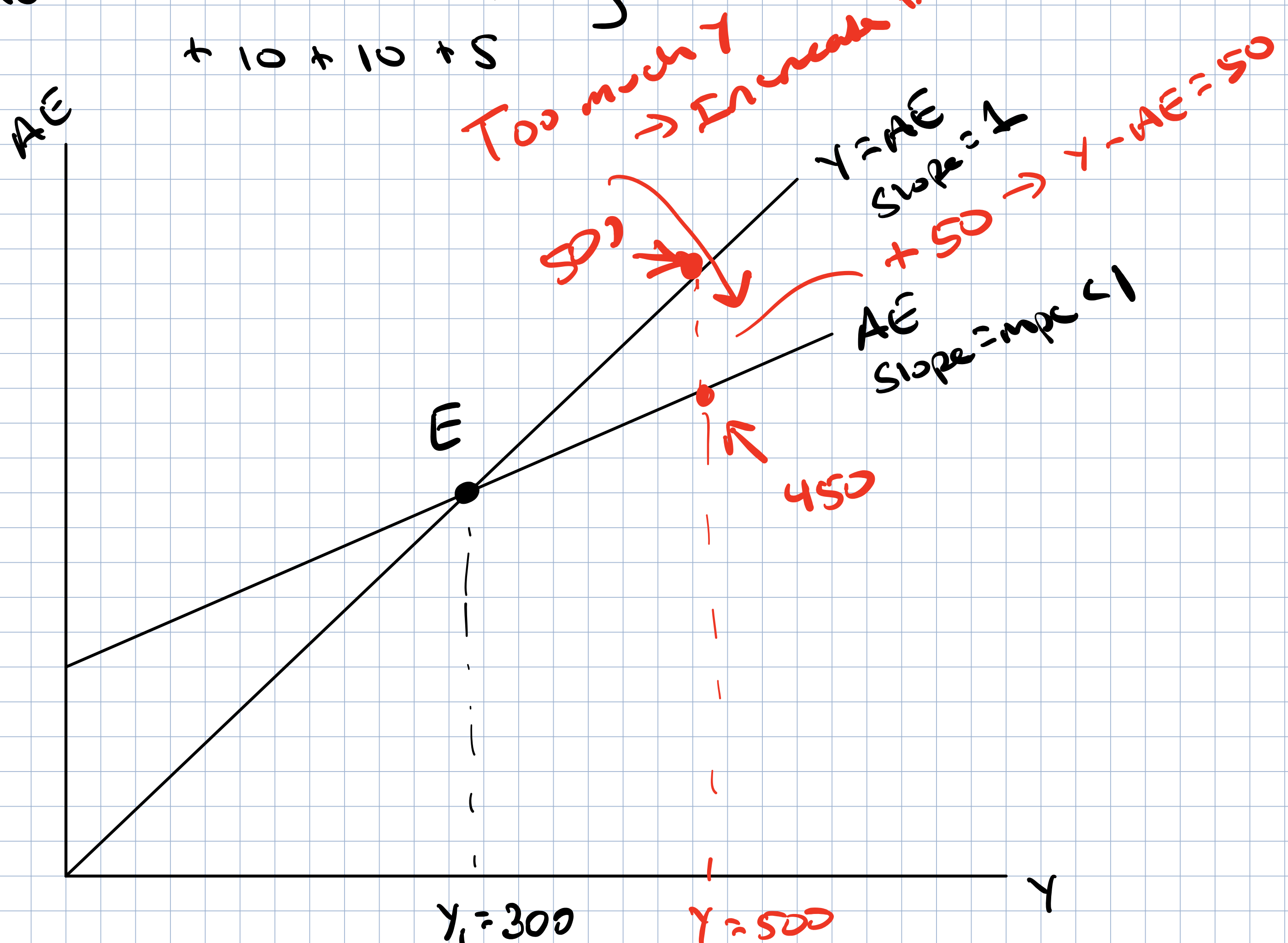
Show Y = AE in a table

Y	C	+ I^P	+ G	+ Nx	= AE	Y − AE Chg. in Inventory
200	200	10	10	5	225	−25
300	275	10	10	5	300	∅
400	350	10	10	5	375	400 − 375 = +25
500	425	10	10	5	450	500 − 450 = +50
600	500	10	10	5	525	+75

What is the equilibrium level of Y?

$$AE = C + I^P + G + Nx \Big\} C + 25$$

$$+ 10 + 10 + 5$$



T = 100, what is AC?

$$C = AC + mpc(Y - T)$$

$$C = AC + 0.75(Y - 100)$$